Fiscal Year:	FY 2013	**Task Last Updated:**	FY 11/08/2013
PI Name:	Oman, Charles M. Ph.D.		
Project Title:	Validation of Assessment Tests and Countermeasures for Detecting and Mitigating Changes in Cognitive Function During Robotics Operations		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Sensorimotor Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	07/01/2009	**End Date:**	07/31/2013
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2013 per NSBRI (Ed., 7/2/13) NOTE: change in Team to Sensorimotor Adaptation Team, from Neurobehavioral and Psychosocial Factors Team, per NSBRI report (Ed., 8/8/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Czeisler, Charles (Brigham and Women's Hospital) Liu, Andrew (Massachusetts Institute of Technology) Lockley, Steven (Brigham and Women's Hospital) Barger, Laura (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-NBPF02001		
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Performance Goal Text:			

Task Description:	Space telerobotics operations aboard the Shuttle and ISS have been successful to date - but not entirely without incident. Fatigue is common among robotics operators due to the long (6-8 hours) duration robotic operations, chronic sleep deprivation, and abrupt "slam shifts" in work schedule, causing circadian misalignment. The impairment of cognitive abilities has been identified as a factor leading to errors in performance. Countermeasures are necessary to improve crew safety and efficiency. The ISS has recently been outfitted with Solid State Lighting Modules, which permit the use of blue enriched light. This research project involves a collaboration between the MIT Man Vehicle Laboratory (MVL), and the Division of Sleep Medicine at Brigham and Women's Hospital (BWH), with the assistance of collaborators at Jefferson Medical College, NASA Kennedy Space Center, and the NASA Johnson Space Center Astronaut Office and Mechanical and Robotics Systems Group. The goal has been to: (1) characterize the changes in performance and mental workload during simulated robotic operations, (2) validate proxy cognitive and drowsiness assessment tests as predictors of performance changes in a complex operational task, and (3) test the efficacy of fatigue countermeasures (e.g., blue enriched white light, caffeine) to improve cognition during robotic operations. Initial experiments (Forman 2011, Lowenthal, 2012) to validate robotics training, performance and mental workload paradigms, and to confirm the relationship between spatial ability tests and performance. Then, over a 20 month period, subjects were screened for robotics aptitude and medical factors, and followed a normal 8 hour/night for several weeks while training on a battery simulated space telerobotics tasks. Then they underwent a week of 6 hour per night sleep restriction, and entered a time-cue-free laboratory environment, where they were tested over a 12 day period, before and after repeated 9 hour slam shifts in their sleep-work schedules. 16 completed the study. Robotics tasks included ISS-like fly-to/grapple, autosequence monitoring, and track-and-capture manual control tasks. An imbedded secondary visual detection task was used to assess spare attention and thus mental workload. Performance on proxy cognitive (Number-Letter, Digit Symbol Substitution), and psychomotor (10 min PVT) tasks is evaluated at intervals. By the end of the 6 hour robotics session, subjects had been awake for 18 hours. Physiological measures (plasma and saliva melatonin and EEG) and subjective sleepiness (KSS, KDT) and alertness measures were taken throughout. After each slam shift, subjects were allowed 8 hours of recovery sleep on the pre-slam schedule. The first slam shift performance study was performed in 90 lux white light, and served as a control. In subsequent 3 day cycles, subjects were tested under three treatment conditions, whose order was randomized across subjects: a) 90 lux blue-enriched white light, b) 90 lux white light and hourly caffeine (0.3 mg/kg), and c) 90 lux blue-enriched light and caffeine. Experiments concluded in January, 2013, and a blinded analysis of the robotics, proxy cognitive, and physiological data is underway. A preliminary analysis (Galvan, 2012) of control and white light session data confirmed a predictive relationship between spatial ability tests (e.g. Vandenberg MRT) and screening test performance. As expected, there were large inter-subject differences in primary task performances, but most subjects were able to maintain primary performance despite slam shifting. However a time-on-task related effect on secondary task response time was found, confirming our pilot experiment results, and indicating a decrease in attentional reserves towards the end of the lengthy robotic session. Blinded analysis of the robotics data from the countermeasures session is now complete. Analyses of KSS, KDT, melatonin, and EEG analyses are now in the final stages. Our protocol requires blinded statistical analysis till our entire data set is finalized, but we have been encouraged that mixed hierarchical regression models for secondary task performance for various robotics task types has shown significant and consistent countermeasure treatment effects as well as the expected time-on-task effects. The pilot experiments and robotics analysis have resulted in three MIT Master's theses (Forman, Lowenthal, Galvan). Also, Master thesis study bimanual cross-coupling in telerobotics (Wang, 2011) was completed, as detailed in our 2012 (Year 3) report.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although monochromatic blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results have not been tested for broadband blue-enriched white light or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the effect on complex performance, such as robotics performance. Similarly, while caffeine use is widespread, including on the ISS, uncontrolled use of caffeine may not be optimally timed or of the correct dose to alleviate sleepiness on duty and may interfere with subsequent sleep, thereby increasing fatigue the next day. The current study will investigate the effects of light and caffeine as fatigue countermeasures in a controlled environment with the long-term view to developing specialized countermeasure schedules to maximize alertness and performance of space crew in an environment where even small fatigue-related errors could have catastrophic consequences. These results will also have widespread application on Earth. Approximately 18,000 Air Force pilots and flight crew are exposed to conflicting circadian signals and resulting fatigue from long flights. Tens of thousands of troops are subjected to rapid deployment world-wide and are often required to maintain alertness while suffering from jet lag. Additionally, nearly 180,000 troops are required to be vigilant for more than 24 hours at a time, and thousands of Navy and Coast Guard personnel work for extended periods under circadian misalignment. Over 200,000 commercial airline pilots, flight crews, and air traffic controllers are also subjected to extended shifts and circadian misalignment. The NTSB implicated hours of duty and fatigue issues in the 2009 Colgan Air commuter airline crash in Buffalo, NY, and this year the FAA has overhauled rules for air traffic controllers after several highly publicized issues where controllers apparently fell asleep. More than 230,000 ER doctors, staff, and medical residents work extended shifts for 30 hours or more. Approximately 1.9 million public safety personnel (police, corrections officers, firefighters, etc.) work extended shifts and are exposed to emergencies which require vigilance for long periods. In total, more than 3 million Americans would directly benefit from the findings of this research. In addition, applications may be possible in standard occupational settings to improve general day-time alertness; in educational settings such as in training sessions and conferences, colleges, and schools where enhanced alertness would assist learning and memory; as a fatigue countermeasure for sleepy car and truck drivers in order to provide them with a short bursts of alertness that will allow them to find a safe place to stop driving and take a break. These wider applications have the potential to reach tens of millions of customers and indeed anyone who could benefit from improved alertness and performance.

Task Progress:	Specific aims are to: (1) characterize the changes in performance and mental workload during simulated robotic operations, (2) validate proxy cognitive and drowsiness assessment tests as predictors of performance changes in a complex operational task, and (3) test the efficacy of fatigue countermeasures (e.g., blue enriched white light, caffeine) to improve cognition during robotic operations. Initial experiments (Forman 2011, Lowenthal, 2012) were to validate robotics training, performance, and mental workload paradigms, and to confirm the relationship between spatial ability tests and performance. Then, over a 20 month period subjects were screened for robotics aptitude and medical factors, and followed an normal 8 hour/night for several weeks while training on a battery simulated space telerobotics tasks. Then they underwent a week of 6 hour per night sleep restriction, and entered a time-cue-free laboratory environment, where they were tested over a 12 day period, before and after repeated 9 hour slam shifts in their sleep-work schedules. 16 subjects completed the study. Robotics tasks included ISS-like fly-to/grapple, autosequence monitoring, and track-and-capture manual control tasks. An imbedded secondary visual detection task was used to assess spare attention and thus mental workload. Performance on proxy cognitive (Number-Letter, Digit Symbol Substitution), and psychomotor (10 min PVT) tasks is evaluated at intervals. By the end of the 6 hour robotics session, subjects had been awake for 18 hours. Physiological measures (plasma and saliva melatonin and EEG) and subjective sleepiness (KSS, KDT) and alertness measures were taken throughout. After each slam shift, subjects were allowed 8 hours of recovery sleep on the pre-slam schedule. The first slam shift performance study was performed in 90 lux white light, and served as a control. In subsequent 3 day cycles, subjects were tested under three treatment conditions, whose order was randomized across subjects: a) 90 lux blue-enriched white light, b) 90 lux white light and hourly caffeine (0.3 mg/kg), and c) 90 lux blue-enriched light and caffeine. Experiments concluded in January, 2013, and a blinded analysis of the robotics, proxy cognitive, and physiological (melatonin and EEG) data is underway.
Bibliography Type:	Description: (Last Updated: 12/18/2024)
Abstracts for Journals and Proceedings	Galvan R, Liu A, Rueger M, Flynn-Evans E, Natapoff A, Lockley S, Oman C. "Validation of assessment tests and countermeasures for detecting and mitigating changes in cognitive function due to fatigue during telerobotic operations." ASMA Panel Session: Sensorimotor Evalutation and Countermeasures for Space Exploration: Moderators C.Oman and A. Mulavara. 84th Annual Scientific Meeting, Aerospace Medical Association, Chicago, IL, May 12-16, 2013. Aviation, Space, and Environmental Medicine. 2013 Apr;84(4):413. , Apr-2013
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Awards	Galvan R. "NASA NSTRF Graduate Fellowship, July 2013." Jul-2013
Dissertations and Theses	Galvan R. "Effects of fatigue on simulated space telerobotics performance: a preliminary study analysis." Master of Science dissertation, Department of Aeronautics and Astronautics, MIT, Cambridge MA, August 23, 2012. 170 pages. http://hdl.handle.net/1721.1/77110 , Aug-2012